VIGNESH KUMAR S

Platform Engineer

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• GitHub Projects: https://github.com/Vignesh-Kumar-svk

Summary

Platform Engineer with 2 years of production experience building cloud automation systems, network provisioning pipelines, and high-availability infrastructure. Reduced VM provisioning turnaround from several hours to under 10 minutes through distributed workflow orchestration using Spring Boot, Camunda BPMN, RabbitMQ, and OpenStack. Hands-on expertise in Fortinet firewall automation, HAProxy load balancer provisioning, and KVM infrastructure — with a track record of owning systems end-to-end from design to production deployment.

Experience

Tata Communications — Chennai

Platform Engineer — 07/2024 - Present

Cloud Infrastructure Automation & High Availability

- Sole developer of a zero-touch HAProxy load balancer provisioning platform on KVM — designed and implemented the full system from scratch, orchestrating VM spin-up, HAProxy configuration, and Zabbix monitoring registration through a Spring Boot + Camunda BPMN workflow engine, Ansible playbooks, and OpenStack SDKs; reduced manual provisioning effort to zero for the operations team.
- Built customer-triggered VIP provisioning workflows exposed via a Spring Boot REST API — dynamically configured HAProxy backends, frontends, and ACLs through the HAProxy Data Plane REST API; implemented Keepalived VRRP active-passive failover with TCP/HTTP load-balancing modes and continuous backend health monitoring, achieving production-grade high availability for enterprise B2B tenants.

Fortinet Firewall & Network Security Automation

- Engineered a Java SSH-based provisioning engine for Fortinet firewall constructs (inter-VLAN routing, static routes, firewall policy management) — optimized from 15s to 4ms per operation (99.97% reduction) via persistent VDOM session pooling and multiplexed SSH channels, eliminating per-request connection overhead across multi-tenant environments.
- Automated OpenStack security group provisioning using Python and OpenStack SDKs — synchronized security group rules with Fortinet firewall policies via a Spring Boot orchestration layer, enforcing consistent VM-level and zone-level access control across multi-tenant B2B cloud environments and eliminating manual policy mismatches.

OpenStack VM provisioning and Automation

- Designed and implemented end-to-end KVM VM provisioning workflows within a Spring Boot + Camunda BPMN orchestration system — built custom Python modules to manage VM lifecycle operations (create, resize, terminate) with full auditability, handling dynamic compute, network, and storage allocation via OpenStack Nova, Neutron, and Cinder APIs.
- Integrated RabbitMQ-based async messaging across the provisioning microservices pipeline — decoupled service communication from synchronous OpenStack API calls, enabling parallel workflow execution and reducing VM provisioning turnaround from several hours to under 10 minutes.

Tata Communications — Chennai

Platform Engineer Intern — 01/2024 - 07/2024

- Integrated Cisco APIC REST APIs into Camunda BPMN workflows to automate end-to-end CRUD operations on Cisco ACI fabric — eliminated manual SDN provisioning steps for cross-functional teams, reducing configuration turnaround for network ops.
- Automated provisioning of core ACI network constructs — Tenant Profiles, Bridge Domains, EPGs, EPG-to-BD bindings, and AAEP profile groupings — via programmatic APIC REST API calls, cutting per-construct configuration time from manual hours to seconds.

Skills

Languages & frameworks: Java, Spring Boot, Python, Angular, MySQL

Infrastructure & cloud: OpenStack (Nova, Neutron, Cinder), KVM, HAProxy, Docker, Linux, AWS (EC2, S3, Lambda, IAM, SQS)

Networking, messaging & observability: Fortinet (FortiOS), RabbitMQ, Kafka, Zabbix, Prometheus, Grafana

Education

Rajalakshmi Institute of Technology

Computer Science and Engineering (CSE) | CGPA: 8.59/10 — 06/2020 - 06/2024